

一、插件化开发——事件监听与模块扩展

结果：

1. 使用插件化开发模式创建一个插件，保证插件能够正常加载（ctp.log中搜索插件id可找到记录）（4分）

```
15:31:41.408,,,,[main],[INFO],PluginSystemInit: - Load Plugin : 20200811, supervisionAdvanced, 督查督办高级插件
15:31:41.409,,,,[main],[INFO],PluginSystemInit: - start analyze plugin:taskmanage
15:31:41.410,,,,[main],[INFO],PluginSystemInit: - Load Plugin : 30001, taskmanage, 工作任务
15:31:41.411,,,,[main],[INFO],PluginSystemInit: - start analyze plugin:test408
15:31:41.413,,,,[main],[INFO],PluginSystemInit: - Load Plugin : 20001, test408, 插件测试
15:31:41.413,,,,[main],[INFO],PluginSystemInit: - start analyze plugin:test408sso
15:31:41.415,,,,[main],[INFO],PluginSystemInit: - Load Plugin : 40802, test408sso, 单点登录
15:31:41.416,,,,[main],[INFO],PluginSystemInit: - start analyze plugin:timeView
15:31:41.416,,,,[main],[INFO],PluginSystemInit: - Load Plugin : 35003, timeView, 时间视图
15:31:41.417,,,,[main],[INFO],PluginSystemInit: - start analyze plugin:timezone
```

2. 在项目中添加事件监听组件，编写代码实现：

- 监听组织模型的人员创建事件（AddMemberEvent），获取被创建人员的姓名和部门信息，调用系统的消息接口给创建人发送欢迎入职的消息（userMessageManager.sendSystemMessage）（3分）



- 监听流程结束后事件，记录流程 ID 和审批完成时间、流程标题到ctp.log日志中（CtpLogFactory.getLog）（3分）

```
18:15:41.488,2qYFMYQ,,,[H-26],[INFO],ColWorkFlowEventListener: - onWorkItemFinished.start
18:15:41.488,2qYFMYQ,,,[H-26],[INFO],ColWorkFlowEventListener: -
laiqiao,workfloweventliserter.onWorkitemFinished return!
18:15:41.492,2qYFMYQ,,,[H-26],[INFO],ColWorkFlowEventListener: - onActivityFinished.start
18:15:41.498,2qYFMYQ,,,[H-26],[INFO],ColWorkFlowEventListener: - onProcessFinished.start
18:15:41.591,2qYFMYQ,,,[H-26],[INFO],CollaborationExtIntegrationUtil: - 推送第三方的summary, summaryID=
2882565827405303488,summaryState=3
18:15:41.591,2qYFMYQ,,,[H-26],[INFO],ColWorkFlowEventListener: - 流程结束触发表单高级事件: affair.getsubject=测
试;1743794637836078493;summaryID=2882565827405303488
18:15:41.591,2qYFMYQ,,,[H-26],[INFO],ColWorkFlowEventListener: - 发送事件: 9
18:15:41.612,2qYFMYQ,,,[H-26],[INFO],ColWorkFlowEventListener: - 赖
桥,onProcessFinished,colSummary.id:2882565827405303488
18:15:41.641,2qYFMYQ,,,[H-26],[INFO],ColWorkFlowEventListener: - onProcessWorkItemAfter.start
18:15:41.641,2qYFMYQ,,,[H-26],[INFO],CollaborationSubmitWorkflowNormalListener: - 协同提交工作流完成
ColHandleType[finish],result[{"caselId":"3734707011456027906","processId":"-688989700782600268","nextMembers"
:"","nextMembersWithoutPolicyInfo":"","hasSubprocess":null,"policyIds":"collaboration","startDR":null,"deadlineDateti
me":null,"businessData":null,"processLogDetails":null,"currentNodeLast":null,"currentActivityChangedId":null,"workflo
wAutoActionEvent":null,"page":null,"startFinished":true}]
```

代码:

```
// 客开: laiqiao [人员创建后给当前创建人发送欢迎入职消息，同步以便取到当前用户] 2026-09-24
0 个用法
🔍
@ListenEvent(event = AddMemberEvent.class, async = false)
public void onAddMember(AddMemberEvent event) {
    try {
        V3xOrgMember member = getMember(event);
        if (member == null) {
            LOG.warn("人员创建事件没有人员对象，跳过欢迎消息");
            return;
        }
        User creator = AppContext.getCurrentUser();
        if (creator == null || creator.getId() == null) {
            LOG.warn("人员创建时没有当前用户，跳过欢迎消息，member=" + member.getName());
            return;
        }
        if (member.getId() == null) {
            LOG.warn("新建人员没有ID，跳过欢迎消息，member=" + member.getName());
            return;
        }
        sendWelcomeMessage(member, creator);
    } catch (Exception e) {
        LOG.error("message: 发送欢迎入职消息失败", e);
    }
}

// 客开: laiqiao [流程结束后把流程ID、完成时间和标题写入ctp.log] 2026-09-24
0 个用法
🔍
@ListenEvent(event = CollaborationFinishEvent.class, async = true)
public void onCollaborationFinish(CollaborationFinishEvent event) {
    try {
        if (event == null) {
            LOG.warn("流程结束事件为空");
            return;
        }
        ColSummary summary = event.getSummary();
        CtpAffair affair = event.getAffair();
        String processId = firstNonBlank(summary == null ? null : summary.getProcessId(),
            affair == null ? null : affair.getProcessId());
        String subject = firstNonBlank(summary == null ? null : summary.getSubject(),
            affair == null ? null : affair.getSubject());
        Date finishDate = summary == null || summary.getFinishDate() == null
            ? new Date() : summary.getFinishDate();
        String finishText = Datetimes.format(finishDate, pattern: "yyyy-MM-dd HH:mm:ss");
        LOG.info("流程结束，processId=" + processId + ", finishDate=" + finishText + ", subject=" + subject + ", summaryId=" + event.getSummaryId());
    } catch (Exception e) {
        LOG.error("message: 记录流程结束日志失败", e);
    }
}

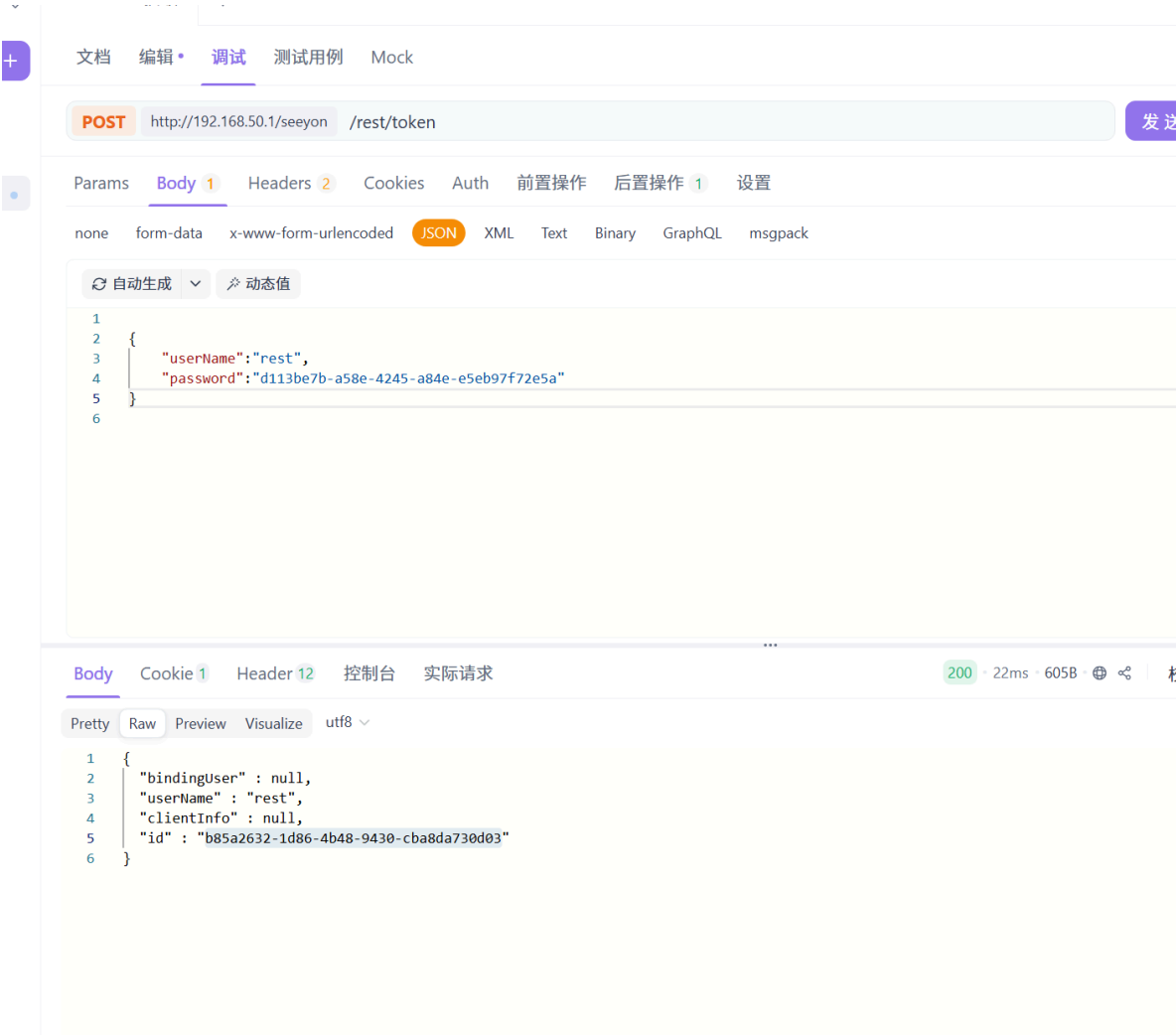
// 客开: laiqiao [安全获取人员创建事件中的人员对象] 2026-09-30
1 个用法
🔍
> private V3xOrgMember getMember(AddMemberEvent event){ return event == null ? null : event.getMember();}

// 客开: laiqiao [发送人员创建欢迎消息] 2026-09-30
1 个用法
```

二、REST 接口——部门人员查询与 Token 调用（15 分）

结果：

- 1. 按照 V5 REST 接口规范（包名 `com.seeyon.ctp.rest.resources`，继承 `BaseResource`），编写部门人员查询接口：
 - 使用 GET 或 POST 方法，返回 JSON 格式数据（6 分）



- 2. 编写客户端调用代码，实现：
 - 通过部门ID（`org_unit`表中`type=Department`）查询部门成员（`orgManager.getMembersByDepartment`），接口返回部门下成员的：人员ID、人员姓名、人员登录名、所在单位名称、所在部门名称、主岗岗位名称、职务名称（5 分）
 - 使用Postman或Apifox等工具成功调用接口（2 分）
- 3. 接口定义必须满足：只使用 GET / POST，新增/修改/删除/查询操作使用对应动作词

(create/update/remove/query) (2 分)

POST token获取 GET 查询部门 + ...

文档 编辑 调试 测试用例 Mock

GET http://192.168.50.1/seeyon/rest/test408/department/member/query 发送 暂停

Params 2 Body Headers Cookies Auth 前置操作 后置操作 1 设置

Query 参数

参数名	参数值	类型	说明
departmentId	= -4351527262217551064	string	
token	= 13f3cd61-1928-4afe-ad9a-c9926b67d11d	string	

添加参数

Body Cookie 1 Header 13 控制台 实际请求 200 · 11ms · 1013B 校验响应 成功

Pretty Raw Preview Visualize JSON utf8

```
4 {
5   "id": -7104128353649802922,
6   "name": "小陈",
7   "loginName": "chen",
8   "accountName": "A单位",
9   "departmentName": "办公室",
10  "postName": "办公室主任",
11  "levelName": "员工"
12 },
13 {
14   "id": -6397809940172356561,
15   "name": "赖桥",
16   "loginName": "laiqiao",
17   "accountName": "A单位",
18   "departmentName": "办公室",
19   "postName": "办公室主任",
20   "levelName": "主任"
21 }
22 1.
```

代码:

2个用法

```
private static final Log LOG = CtpLogFactory.getLog(Test408DepartmentMemberResource.class);
```

// 客开: laiqiao [GET查询本部门人员, 不含子部门] 2026-09-24

Q~

@GET

@Path("department/member/query")

```
public Response queryDepartmentMembers(@QueryParam("departmentId") Long departmentId) {
```

```
    try {
```

```
        OrgManager orgManager = requireOrgManager();
```

```
        requireDepartmentId(departmentId);
```

```
        requireDepartment(orgManager, departmentId);
```

```
        List<Map<String, Object>> result = toMemberItems(orgManager,
```

```
            orgManager.getMembersByDepartment(departmentId, b: true));
```

```
        LOG.info("部门人员查询完成, departmentId=" + departmentId + ", size=" + result.size());
```

```
        return success(result);
```

```
    } catch (IllegalArgumentException e) {
```

```
        return fail(e.getMessage());
```

```
    } catch (Exception e) {
```

```
        LOG.error("message: 部门人员查询失败, departmentId=" + departmentId, e);
```

```
        return fail("查询部门人员失败: " + e.getMessage());
```

```
    }
```

```
}
```

// 客开: laiqiao [获取组织管理器Bean] 2026-09-30

1个用法

Q~

```
private OrgManager requireOrgManager() { return (OrgManager) AppContext.getBean(beanName: "orgManager"); }
```

// 客开: laiqiao [校验部门查询参数] 2026-09-30

1个用法

Q~

```
private void requireDepartmentId(Long departmentId) {
```

```
    if (departmentId == null) {
```

```
        throw new IllegalArgumentException("部门ID不能为空");
```

```
    }
```

```
}
```

// 客开: laiqiao [校验部门类型] 2026-09-30

1个用法

Q~

```
private void requireDepartment(OrgManager orgManager, Long departmentId) throws Exception {
```

```
    V3xOrgDepartment department = orgManager.getDepartmentById(departmentId);
```

```
    if (department == null || department.getType() != OrgConstants.UnitType.Department) {
```

```
        throw new IllegalArgumentException("部门不存在, 或org_unit.type不是Department");
```

```
    }
```

```
}
```

// 客开: laiqiao [转换部门直属人员列表] 2026-09-30

1个用法

Q~

```
private List<Map<String, Object>> toMemberItems(OrgManager orgManager,
```

```
    List<V3xOrgMember> members) throws Exception {
```

```
    List<Map<String, Object>> result = new ArrayList<>();
```

```
    for (V3xOrgMember member : members == null ? Collections.emptyList() : members) {
```

```
        if (member != null) {
```

三、单点登录——Ticket 模式 SSO 实现

结果：

1. 按照V5插件化开发规范，定义单点登录插件，插件能够正常加载（ctp.log中搜索插件id可找到记录）（2分）

```
15:31:41.408,,,,[main],[INFO],PluginSystemInit: - Load Plugin : 20200811, supervisionAdvanced, 督查督办高级插件
15:31:41.409,,,,[main],[INFO],PluginSystemInit: - start analyze plugin:taskmanage
15:31:41.410,,,,[main],[INFO],PluginSystemInit: - Load Plugin : 30001, taskmanage, 工作任务
15:31:41.411,,,,[main],[INFO],PluginSystemInit: - start analyze plugin:test408
15:31:41.413,,,,[main],[INFO],PluginSystemInit: - Load Plugin : 20001, test408, 插件测试
15:31:41.413,,,,[main],[INFO],PluginSystemInit: - start analyze plugin:test408sso
15:31:41.415,,,,[main],[INFO],PluginSystemInit: - Load Plugin : 40802, test408sso, 单点登录
15:31:41.416,,,,[main],[INFO],PluginSystemInit: - start analyze plugin:timeView
15:31:41.416,,,,[main],[INFO],PluginSystemInit: - Load Plugin : 35003, timeView, 时间视图
15:31:41.417,,,,[main],[INFO],PluginSystemInit: - start analyze plugin:timezone
```

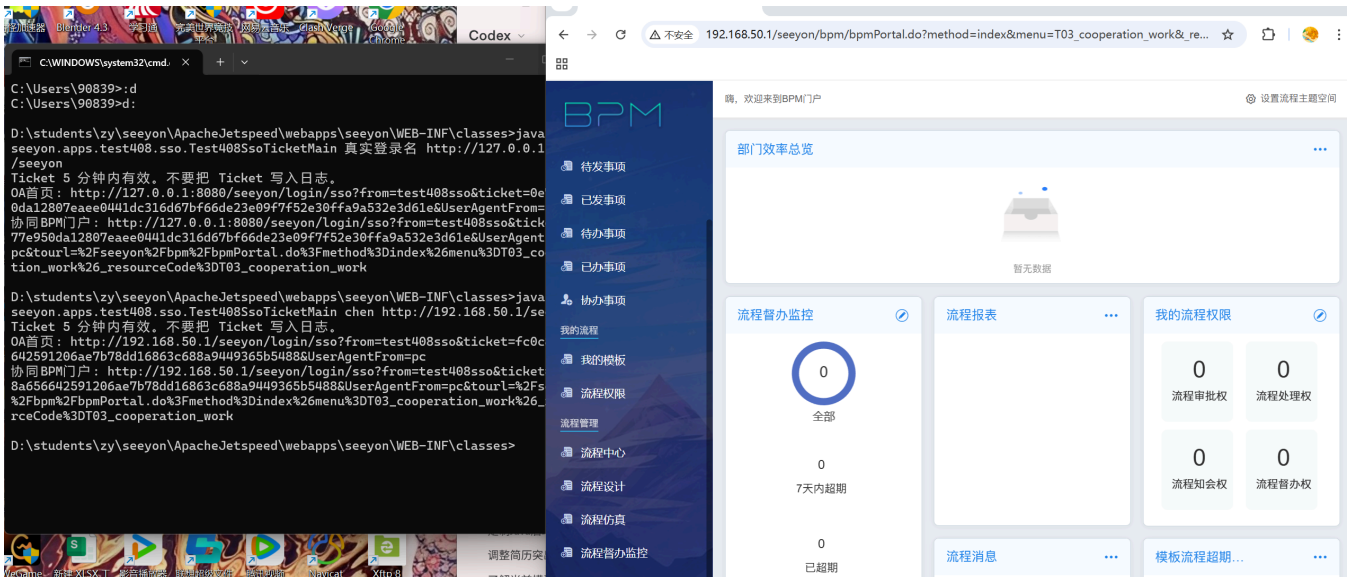
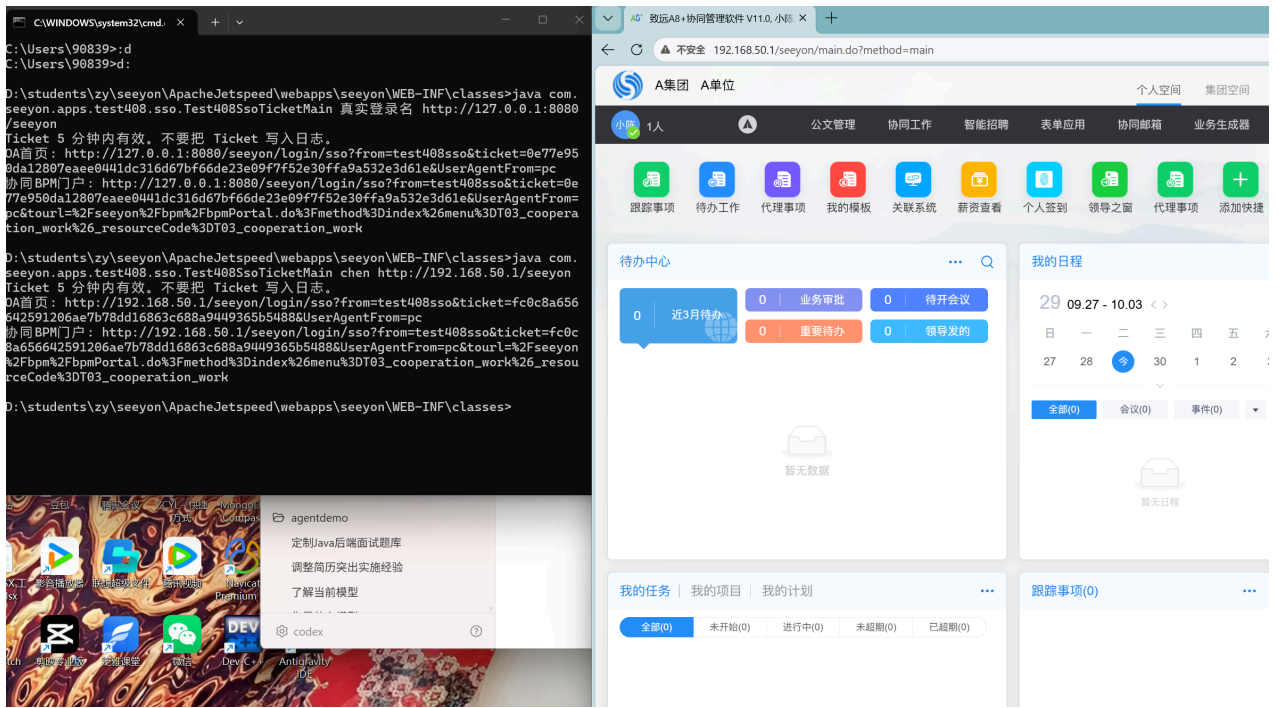
```
交额度=24,已使用量=24,预占用量=0,统一剩余额度=99976,是否降级=false,订单数量=1,待结算数量=0
:30:47.929,,,,[CTPQuartzScheduler SmartThreadPool Worker 5],[INFO],DefaultLicenseRuntimeSyncCoordinator: -
本地流量管理】跳过运行态同步, 原因=运行态无需同步, 运行态=授权ID=MAPSER...RDER,版本号=49,同步状态=CLEAN
用额度=99976,已提交额度=24,已使用量=24,预占用量=0,统一剩余额度=99976,是否降级=false,订单数量=1,待结算数量=
:30:47.929,,,,[CTPQuartzScheduler SmartThreadPool Worker 5],[INFO],LicenseRuntimeSyncQuartzJob: - 【本地流
管理】运行态同步 Quartz 执行完成, 是否执行=false, 是否成功=true, 说明=当前运行态已为 CLEAN, 跳过同步
:30:51.568,2vaaPA9,,,,[H-73],[INFO],Test408SsoHandshake: - SSO 登录成功, loginName=chen,
memberId= 71041128252640802022
```

2. 编写 SSO 验证接口（继承SSOLoginHandshakeAbstract，重写handshake）的核心代码，实现：
 - 解析 ticket 获取用户标识：ticket 需传递加密后的用户登录名，在handshake中解密ticket获取用户登录名（4分）
 - 通过组织模型 API 查找用户（orgManager.getMemberByLoginName）（4分）

```
@Override
public String handshake(String ticket) {
    if (Strings.isBlank(ticket)) {
        return null;
    }
    try {
        String plain = new DesUtils().decrypt(ticket);
        String loginName = parseLoginName(plain);
        if (Strings.isBlank(loginName)) {
            LOG.warn("SSO 登录失败, Ticket 内容无效或已过期");
            return null;
        }
        if (orgManager == null) {
            LOG.error("SSO 登录失败, orgManager 未注入");
            return null;
        }
        V3xOrgMember member = orgManager.getMemberByLoginName(loginName);
        if (member == null || !member.isValid()) {
            LOG.warn("SSO 登录失败, V5 中不存在有效人员: " + loginName);
            return null;
        }
        LOG.info("SSO 登录成功, loginName=" + loginName + ", memberId=" + member.getId());
        return member.getLoginName();
    } catch (Exception e) {
        LOG.error("message: SSO Ticket 握手失败", e);
        return null;
    }
}
```

3. 补充必要的 Spring 配置和握手类代码，自己使用 SSO 接口成功登录 OA 首页，并能跳转到"协同BPM门户"页

面



代码:

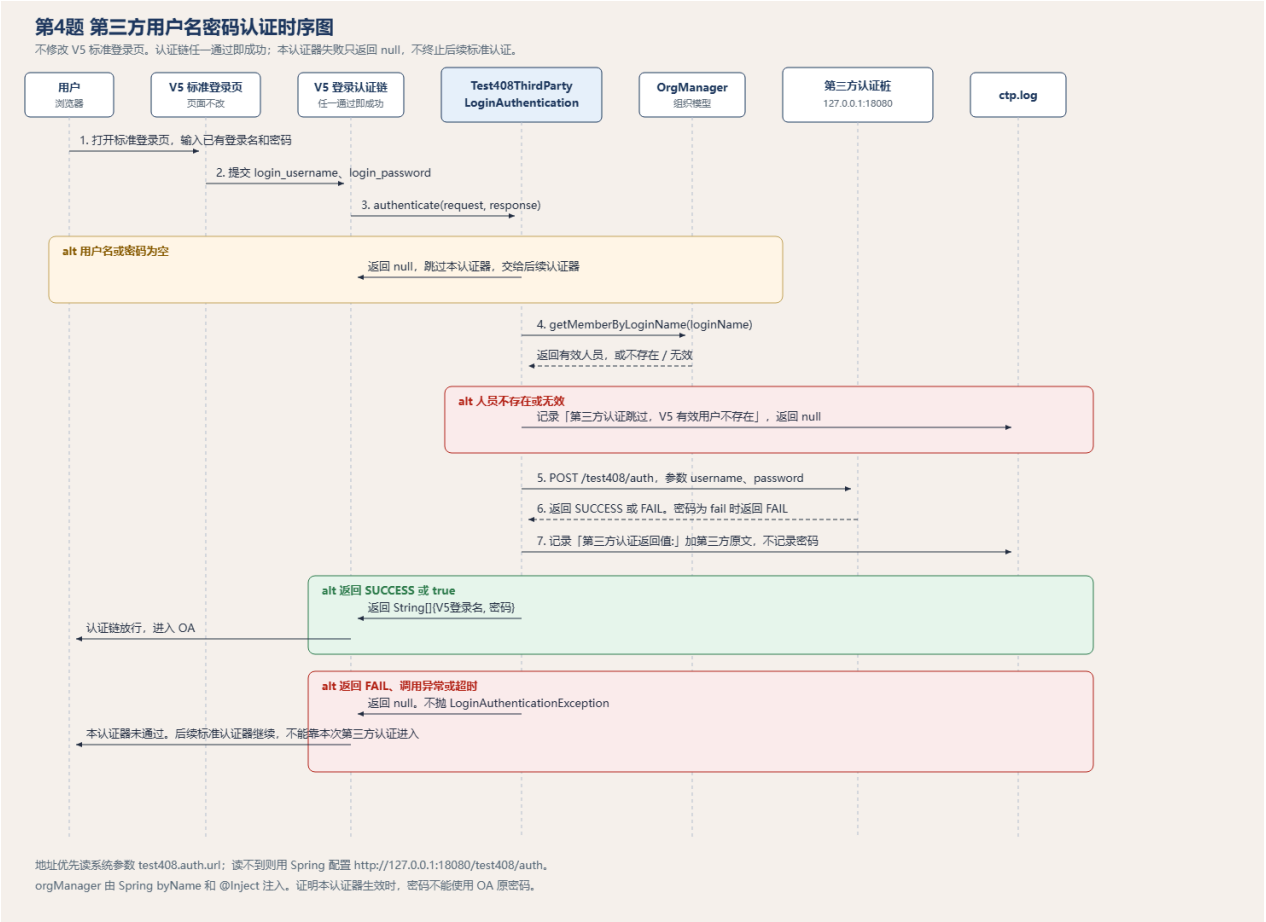
```
25
26 // 客开: laiqiao [解密 Ticket 并按登录名查询 V5 用户] 2026-09-24
27 @Override
28 public String handshake(String ticket) {
29     if (Strings.isBlank(ticket)) {
30         return null;
31     }
32     try {
33         String plain = new DesUtils().decrypt(ticket);
34         String loginName = parseLoginName(plain);
35         if (Strings.isBlank(loginName)) {
36             LOG.warn("SSO 登录失败, Ticket 内容无效或已过期");
37             return null;
38         }
39         if (orgManager == null) {
40             LOG.error("SSO 登录失败, orgManager 未注入");
41             return null;
42         }
43         V3xOrgMember member = orgManager.getMemberByLoginName(loginName);
44         if (member == null || !member.isValid()) {
45             LOG.warn("SSO 登录失败, V5 中不存在有效人员: " + loginName);
46             return null;
47         }
48         LOG.info("SSO 登录成功, loginName=" + loginName + ", memberId=" + member.getId());
49         return member.getLoginName();
50     } catch (Exception e) {
51         LOG.error("message: \"SSO Ticket 握手失败\", e);
52         return null;
53     }
54 }
55
56 // 客开: laiqiao [校验 Ticket 过期时间, 明文格式 loginName%%expireMillis] 2026-09-28
57 1 个用法
58 @ private String parseLoginName(String plain) {
59     if (Strings.isBlank(plain) || plain.indexOf(SEPARATOR) < 0) {
60         return null;
61     }
62     int splitAt = plain.indexOf(SEPARATOR);
63     String loginName = plain.substring(0, splitAt).trim();
64     String expireText = plain.substring(beginIndex: splitAt + SEPARATOR.length()).trim();
65     if (Strings.isBlank(loginName) || Strings.isBlank(expireText)) {
66         return null;
67     }
68     long expireAt;
69     try {
70         expireAt = Long.parseLong(expireText);
71     } catch (NumberFormatException e) {
72         return null;
73     }
74     long now = System.currentTimeMillis();
75     if (expireAt < now || expireAt - now > MAX_TICKET_AGE_MILLIS) {
76         return null;
77     }
78     return loginName;
79 }
```

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <!DOCTYPE beans PUBLIC "-//SPRING//DTD BEAN//EN" "http://www.springframework.org/dtd/spring-beans.dtd">
3 <beans default-autowire="byName">
4     <!-- 客开: laiqiao [注册 Ticket SSO 握手类的 orgManager 由 byName 和 @Inject 注入] 2026-09-28 -->
5     <bean id="test408Sso" class="com.seeyon.ctp.portal.sso.SSOLoginContext">
6         <property name="name" value="test408sso" />
7         <property name="forward" value="true" />
8         <property name="handshake">
9             <bean class="com.seeyon.apps.test408.sso.Test408SsoHandshake" autowire="byName" />
10        </property>
11    </bean>
12 </beans>
```

四、自定义登录认证——第三方认证登录（10 分）

结果

1. 设计第三方认证登录的认证流程，并画出时序图（或文字描述关键步骤）。（2 分）



2. 编写认证类，实现对应的基类（LoginAuthentication），实现：

- 获取用户前端输入的用户名+密码，调用组织模型 API 判断用户是否存在（2 分）
- 调用第三方接口进行密码校验（2 分）
- 成功认证登录（2 分）

3. 正确记录日志到ctp.log中（记录第三方接口的返回值）（2 分）

```
Microsoft Windows [版本 10.0.26100.6584]
(c) Microsoft Corporation. 保留所有权利。

C:\Users\90839>cd D:\students\zy\seeyon\ApacheJetspeed\webapps\seeyon\WEB-INF\classes

D:\students\zy\seeyon\ApacheJetspeed\webapps\seeyon\WEB-INF\classes>java com.seeyon.apps.test408.Login.Test408ThirdParty
AuthServer
第三方认证桩已启动: http://127.0.0.1:18080/test408/auth
任意非空密码返回 SUCCESS; 密码 fail 返回 FAIL。Ctrl+C 停止。
```

```
group=com.seeyon.ctp.common.authenticate.domain.IdentificationDogManager,name=UserIdOrLanAccessMobile,loader=
10:49:43.410,2vfbRec,,,[H-13],[INFO],GroupCacheableFactory: - [createSet]
group=com.seeyon.ctp.common.authenticate.domain.IdentificationDogManager,name=NoChekcIPs,loader=
10:49:43.417,2vfbRec,,,[H-13],[INFO],AppContext: - [trace] plugin spsso checked result true
10:49:43.418,2vfbRec,,,[H-13],[INFO],PrincipalManagerImpl: - authenticate, result false, loginName:jiangge
10:49:43.472,2vfbRec,,,[H-13],[INFO],Test408ThirdPartyLoginAuthentication: - 第三方认证返回值: SUCCESS
10:49:43.473,2vfbRec,,,[H-13],[INFO],Test408ThirdPartyLoginAuthentication: - 第三方认证返回值: SUCCESS
```

【管理】运行态同步 Quartz 执行完成，是否执行=false，是否成功=true，说明=当前运行态已为 CLEAN，跳过同步
0:51:08.433,2vfwld,,,[H-86],[ERROR],LoginMobileSmsAuthentication: - 手机号自动登录，userAgentFrom为空!
0:51:08.434,2vfwld,,,[H-86],[INFO],PrincipalManagerImpl: - authenticate, result false, loginName:jiangge
0:51:08.436,2vfwld,,,[H-86],[INFO],Test408ThirdPartyLoginAuthentication: - 第三方认证返回值: FAIL
0:51:08.436,2vfwld,,,[H-86],[INFO],Test408ThirdPartyLoginAuthentication: - 第三方认证未通过: jiangge
0:51:14.712,,,,[SY-Executors-default-schedule-thread-1],[ERROR],BllIntegrationManagerImpl: - 协同驾驶舱服务不可用,
检查配置 [code : 2055579080]

代码：

```
44
45 // 客开: laiqiao [读取前端登录参数，先校验 V5 人员存在性，再调用第三方认证] 2026-09-24
46 @Override
47 public String[] authenticate(HttpServletRequest request, HttpServletResponse response)
48     throws LoginAuthenticationException {
49     String username = request.getParameter(LoginConstants.USERNAME);
50     String password = request.getParameter(LoginConstants.PASSWORD);
51     if (StringUtils.isBlank(username) || StringUtils.isBlank(password)) {
52         return null;
53     }
54     String loginName = username.trim();
55     try {
56         if (orgManager == null) {
57             LOG.error("第三方认证跳过，orgManager 未注入");
58             return null;
59         }
60         V3xOrgMember member = orgManager.getMemberByLoginName(loginName);
61         if (member == null || !member.isValid()) {
62             LOG.info("第三方认证跳过，V5 有效用户不存在: " + loginName);
63             return null;
64         }
65         String result = verifyByThirdParty(loginName, password);
66         LOG.info("第三方认证返回值: " + result);
67         if (isSuccess(result)) {
68             return new String[] {member.getLoginName(), password};
69         }
70         LOG.info("第三方认证未通过: " + loginName);
71     } catch (Exception e) {
72         LOG.error("第三方认证异常，loginName=" + loginName, e);
73     }
74     return null;
75 }
```

// 客开: [laiqiao](#) [调用第三方认证接口] 2026-09-24

1 个用法

Q v

```
private String verifyByThirdParty(String username, String password) throws Exception {
    String endpoint = resolveAuthUrl();
    if (StringUtils.isBlank(endpoint)) {
        throw new IllegalStateException("未配置第三方认证地址 " + AUTH_URL_KEY);
    }

    HttpURLConnection connection = (HttpURLConnection) new URL(endpoint).openConnection();
    connection.setRequestMethod("POST");
    connection.setConnectTimeout(10000);
    connection.setReadTimeout(10000);
    connection.setDoOutput(true);
    connection.setRequestProperty("Content-Type", "application/x-www-form-urlencoded;charset=UTF-8");
    String body = "username=" + URLEncoder.encode(username, "UTF-8")
        + "&password=" + URLEncoder.encode(password, "UTF-8");
    try (OutputStream output = connection.getOutputStream()) {
        output.write(body.getBytes("UTF-8"));
    }

    InputStream input = connection.getResponseCode() >= 400
        ? connection.getErrorStream() : connection.getInputStream();
    return read(input);
}
```

// 客开: [laiqiao](#) [系统参数优先, Spring 配置兜底] 2026-09-28

1 个用法

Q v

```
private String resolveAuthUrl() {
    String configured = SystemProperties.getInstance().getProperty(AUTH_URL_KEY);
    if (StringUtils.isNotBlank(configured)) {
        return configured.trim();
    }

    return authUrl;
}
```

1 个用法

Q v

```
private boolean isSuccess(String result) {
    return "SUCCESS".equalsIgnoreCase(StringUtils.trim(result))
        || "true".equalsIgnoreCase(StringUtils.trim(result));
}
```

1 个用法

Q v

```
private String read(InputStream input) throws Exception {
    if (input == null) {
        return "";
    }

    StringBuilder result = new StringBuilder();
    try (BufferedReader reader = new BufferedReader(new InputStreamReader(input, "UTF-8"))) {
        String line;
        while ((line = reader.readLine()) != null) {
            result.append(line);
        }
    }

    return result.toString();
}
```

}

五、CAP4 自定义控件——超链接控件

结果：

1. 编写控件的 Java 类，继承 FormFieldCustomCtrl，实现：
 - 指定控件js路径、控件key、控件名称（4 分）
2. 编写控件的 js，包含：
 - 在控件编辑态渲染两个输入框：标题、URL（2分）
 - 在控件浏览态渲染超链接按钮（2 分）

3. 控件编辑态输入的标题、URL可正常保存，控件浏览态渲染的超链接可正常点击使用window.open打开URL。
(2 分)

测试(小陈 2026-09-29 18:03)

协同

小陈

2026-09-29 18:04:10

表单

流程

☆

⊕

📄

🖨

测试表单

姓名

确认签名

是否同意

[测试](#)

代码：

```

7
8 // 客开: laiqiao [CAP4超链接自定义控件] 2026-09-29
  2个用法
9 public class Test408HyperlinkCtrl extends FormFieldCustomCtrl {
10
11     1个用法
12     public static final String CTRL_KEY = "4085617293648510001";
13     /** 字段长度上线后不可修改, 数据库列按此长度创建。 */
14     1个用法
15     public static final String FIELD_LENGTH = "500";
16
17     // 客开: laiqiao [控件key, 上线后不可变] 2026-09-29
18     @Override
19     public String getKey() { return CTRL_KEY; }
20
21     // 客开: laiqiao [控件名称走插件国际化] 2026-09-29
22     @Override
23     public String getText() { return ResourceUtil.getString( key: "com.cap.ctrl.test408.hyperlink.text"); }
24
25     // 客开: laiqiao [pluginId必须在init中设置, Spring不要预先注入] 2026-09-29
26     @Override
27     public void init() {
28         this.setPluginId("test408");
29         this.setIcon("cap-icon-lianjie");
30     }
31
32     0个用法
33     @Override
34     public String[] getDefaultVal(String defaultValue) { return new String[0]; }
35
36     // 客开: laiqiao [PC运行态资源。控件注入信息用对象字面里, 命名空间field_加key] 2026-09-29
37     0个用法
38     @Override
39     public String getPCInjectionInfo() {
40         return "{path:'apps_res/cap/customCtrlResources/test408HyperlinkCtrlResources/',"
41             + "jsUri:'js/test408HyperlinkPcRunning.js',"
42             + "initMethod:'init',"
43             + "nameSpace:'field_' + this.getKey() + '"}";
44     }
45
46     // 客开: laiqiao [本考核不要求移动端] 2026-09-29
47     0个用法
48     @Override
49     public String getMBInjectionInfo() { return null; }
50
51     // 客开: laiqiao [字段长度500, 上线后不可改] 2026-09-29
52     0个用法
53     @Override
54     public String getFieldName() { return FIELD_LENGTH; }
55
56     /**
57      * 产品默认 canUse() 是 hasPlugin("cap4") 且 isValid()。
58      * valid 默认 false, 未做云端授权时设计器看不到控件。
59      * 这里设置是为了考核时能使用, 不是云端授权校验。
60      */
61
62

```

六、CAP4 自定义按钮——一键提醒

结果：

- 1. 编写按钮的 Java 类，继承 CommonBtn，实现：
 - 指定控件js路径、控件key、控件名称（3 分）
 - 获取当前表单对象、指定查询的字段对象（2 分）
 - 使用JDBCAGENT查询满足条件的数据条数（2 分）
 - 发送系统消息提醒（1 分）
- 2. 编写控件的 js，点击按钮后调用后端查询数据、发送消息，成功后前端给出反馈（2 分）

```
owner = ["710712035075002022"] 持有锁action = ["1"]
09:44:18.717,2B8Zkf7,,,[H-91],[INFO],CAP4FormResource: - createOrEdit execute use 12 , build result use 2
09:44:19.624,,,[SY-Executors-CTP-schedule-thread-2],[INFO],lockTrace: - 本次清理超时锁耗时【0】毫秒,超时锁列表详情:[]
09:44:23.218,2B90HvY,,,[H-92],[INFO],Test408RemindBtn: - 一键提醒已发送, formId=5794105079035086457, field=field0005, count=2, receiver=8014431074992736158
09:44:24.481,,,[SY-Executors-default-schedule-thread-1],[ERROR],BIIntegrationManagerImpl: - 协同驾驶舱服务不可用, 请检查配置 [code: 2055579080]
```



代码:

```
56 // 客开: laiqiao [按钮参数用button弹窗。text型属性设计器不落库] 2026-09-29
57 @Override
58 public void init() {
59     this.setPluginId("test408RemindBtn");
60     this.setIcon("cap-icon-custom-button");
61     if (this.params != null && !this.params.isEmpty()) {
62         return;
63     }
64     BtnParamDefinition remindConfig = new BtnParamDefinition();
65     remindConfig.setDialogUrl("apps_res/cap/customCtrlResources/test408RemindBtnResources/html/remindConfig.html");
66     remindConfig.setDisplay("com.cap.btn.test408.remind.param.display");
67     remindConfig.setName("remindConfig");
68     remindConfig.setParamType(Enums.BtnParamType.button);
69     remindConfig.setDialogWidth("520");
70     remindConfig.setDialogHeight("260");
71     addDefinition(remindConfig);
72 }
73
74 @Override
75 public String getKey() { return BTN_KEY; }
76
77 // 客开: laiqiao [命名空间必须是customBtn加key] 2026-09-29
78 2 个月注
79 @Override
80 public String getNameSpace() { return "customBtn" + this.getKey(); }
81
82 @Override
83 public String getText() { return ResourceUtil.getString( key: "com.cap.btn.test408.remind.text"); }
84
85 0 个月注
86 @Override
87 public String getPCInjectionInfo() {
88     return "{\n\"path\": \"apps_res/cap/customCtrlResources/test408RemindBtnResources/\", \"\n\"
89         + \"jsUri\": \"js/\" + this.getNameSpace() + \".umd.min.js\", \"\n\"
90         + \"initMethod\": \"init\", \"\n\"
91         + \"nameSpace\": \"\" + this.getNameSpace() + \"\"}";
92 }
93
94
95
96
97
```

```

// 客开: laiqiao [查询字段非空条数并给指定登录名发系统消息] 2026-09-29
1 个用法
🔍
public Map<String, Object> queryAndRemind(Long formId, String fieldKey, String loginName, Long senderId) throws BusinessException {
    validateRequest(formId, fieldKey, loginName, senderId);

    FormBean formBean = requireForm(formId);
    FormFieldBean fieldBean = requireField(formBean, fieldKey);
    String tableName = fieldBean.getOwnerTableName();
    String columnName = fieldBean.getName();
    validateIdentifiers(tableName, columnName);

    int count = queryCount(tableName, columnName, isTextField(fieldBean.getFieldType()));
    V3xOrgMember receiver = requireReceiver(loginName);
    String content = buildMessage(formBean, fieldBean, count);

    sendReminder(senderId, receiver, content);
    LOG.info("一键提醒已发送, formId=" + formId + ", field=" + columnName
        + ", count=" + count + ", receiver=" + receiver.getId());
    return buildResult(formBean, fieldBean, count, loginName, content);
}

// 客开: laiqiao [集中校验一键提醒请求和依赖] 2026-09-30
1 个用法
🔍
private void validateRequest(Long formId, String fieldKey, String loginName, Long senderId)
    throws BusinessException {
    requireValue(formId, message: "formId不能为空");
    requireText(fieldKey, message: "统计字段不能为空");
    requireText(loginName, message: "接收人登录名不能为空");
    requireValue(senderId, message: "发送人不能为空");
    requireDependencies();
}

// 客开: laiqiao [校验按钮运行依赖] 2026-09-30
1 个用法
🔍
private void requireDependencies() throws BusinessException {
    if (cap4FormManager == null || orgManager == null || userMessageManager == null) {
        throw new BusinessException("一键提醒按钮依赖未注入");
    }
}

// 客开: laiqiao [统一校验必填参数] 2026-09-30
2 个用法
🔍
private void requireValue(Object value, String message) throws BusinessException {
    if (value == null) {
        throw new BusinessException(message);
    }
}

// 客开: laiqiao [统一校验文本参数] 2026-09-30
2 个用法
🔍
private void requireText(String value, String message) throws BusinessException {
    if (value == null || value.trim().length() == 0) {
        throw new BusinessException(message);
    }
}

```

七、CIP 集成平台——待办推送、消息推送、单点登录

结果：

1. 用户采用免绑定（手机号匹配）方式，使用第三方推送模式，用 Postman、Apifox等工具调用CIP平台的待办、消息接口，OA 上的用户能够正常接收待办、消息（4 分）



文档

编辑

调试

测试用例

Mock

POST

http://192.168.50.1/seeyon /rest/thirdpartyMessage/receive/singleMessage

Params 1

Body 1

Headers 1

Cookies

Auth

前置操作

后置操作 1

设置

none

form-data

x-www-form-urlencoded

JSON

XML

Text

Binary

GraphQL

msgpack

自动生成



动态值

```
1 {
2   "thirdpartyRegisterCode": "3001",
3   "thirdpartyMessageId": "ERP-MSG-001",
4   "messageContent": "今日ERP待办汇总",
5   "noneBindingSender": "13419289743",
6   "noneBindingReceiver": "17623741028",
7   "messageURL": "https://www.baidu.com",
8   "messageH5URL": "https://www.baidu.com",
9   "creation_date": "2026-09-30 10:00:00"
10 }
```

Body

Cookie 1

Header 12

控制台

实际请求

Pretty

Raw

Preview

Visualize

JSON

utf8



```
1 {
2   "success": true,
3   "errorMsgs": []
4 }
```

>

二

三

四

五

六

29

今

1

2

3

我的消息

全部(5)

@我的(0)

回复我的(0)

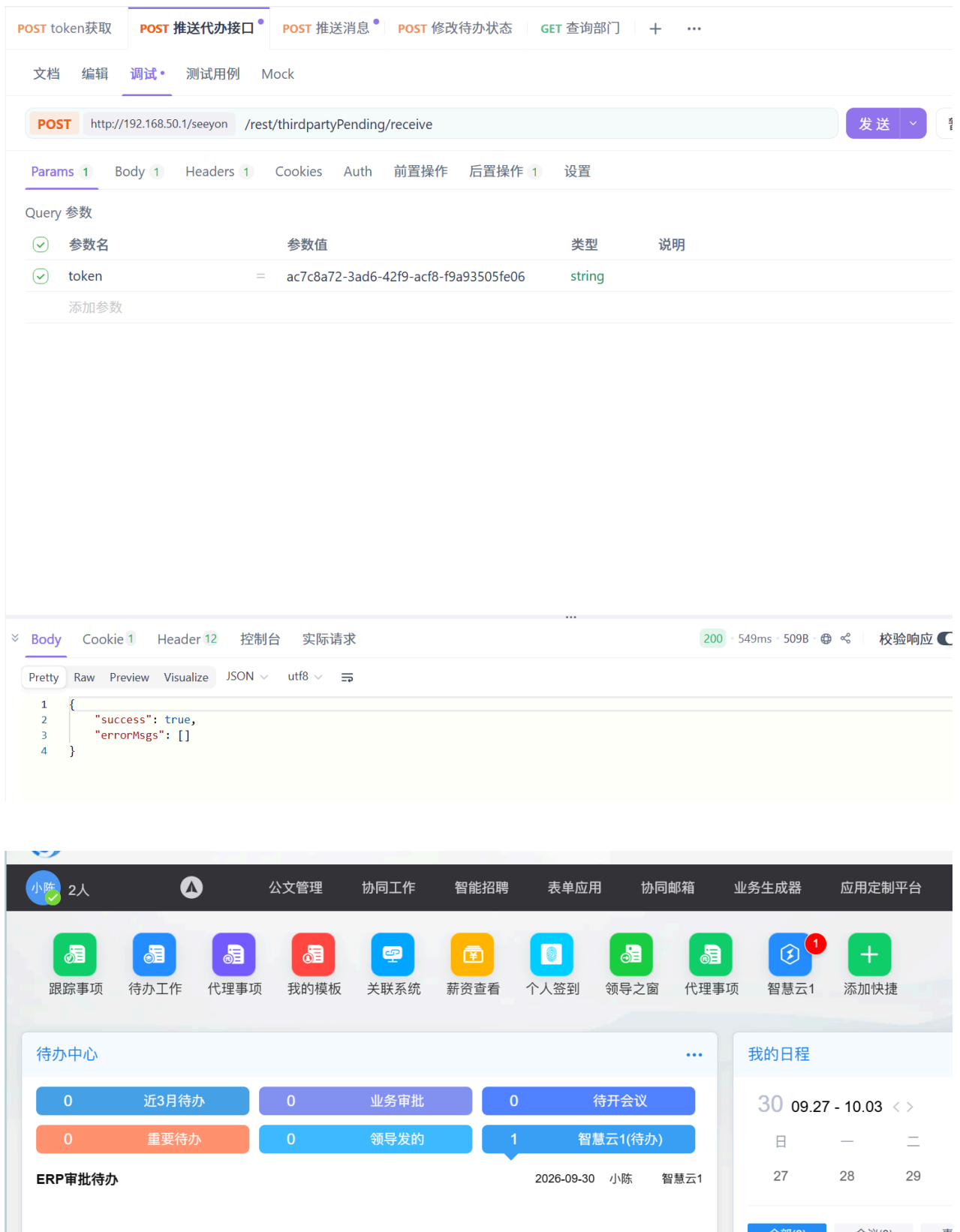


[智慧云1]今日ERP待办汇总

11:54 今天

集成平台消息 小陈

... [unreadable text] ...



2. 编写 SpringBoot 项目，实现 SSO 认证服务：

- 编写 SSO 接口，接收 OA 传递的 v5ticket 并反向调用 OA 的验证接口获取用户信息，将 v5ticket 和用户信息保存到缓存中（3 分）
- 编写登录接口，接收 OA 传递的 v5ticket 参数，在缓存中根据 v5ticket 找到对应用户，将用户信息输入到页面中（3 分）

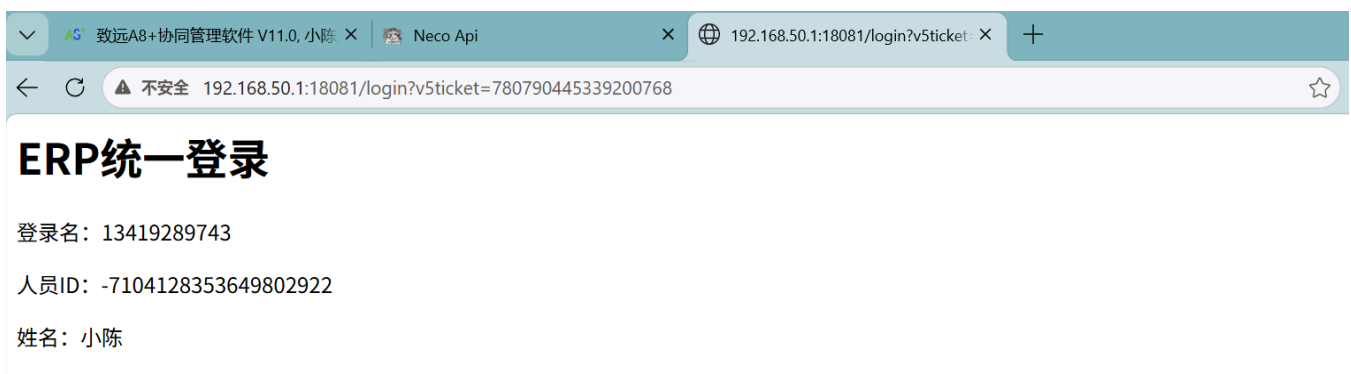
```
4 个用法
@Component
public class OaTicketClient {

    2 个用法
    private final String oaBaseUrl;

    public OaTicketClient(@Value("${test408.oa-base-url:http://127.0.0.1:seevon}") String oaBaseUrl) {
        this.oaBaseUrl = trimTrailingSlash(oaBaseUrl);
    }

    // 容开: laiqiao [读取 OA 响应头中的 LoginName 等用户信息] 2026-09-29
    2 个用法
    @v
    public TicketUser verify(String ticket) throws Exception {
        if (ticket == null || ticket.trim().length() == 0) {
            return null;
        }
        String target = oaBaseUrl + "/thirdpartyController.do?ticket="
            + java.net.URLEncoder.encode(ticket, StandardCharsets.UTF_8.name());
        HttpURLConnection connection = null;
        try {
            connection = (HttpURLConnection) new URL(target).openConnection();
            connection.setInstanceFollowRedirects(false);
            connection.setRequestMethod("GET");
            connection.setConnectTimeout(5000);
            connection.setReadTimeout(5000);
            connection.getResponseCode();
            String loginName = connection.getHeaderField("name: LoginName");
            if (loginName == null || loginName.trim().length() == 0) {
                return null;
            }
            String memberName = connection.getHeaderField("name: MemberName");
            if (memberName != null && memberName.trim().length() > 0) {
                memberName = URLDecoder.decode(memberName, StandardCharsets.UTF_8.name());
            }
            return new TicketUser(loginName, connection.getHeaderField("name: MemberId"), memberName);
        } finally {
            if (connection != null) {
                connection.disconnect();
            }
        }
    }
}
```

3. 配置第三方系统到 OA 首页的磁贴栏目中，点击第三方应用可打开第三方页面，显示用户信息（5 分）



八、编写超级节点——预算校验

1. 自行设计费用报销单据，包含 部门预算档案底表、费用报销申请流程表（5 分）
2. 编写超级节点核心代码，继承 BaseSuperNodeAction，实现

- 获取表单中的报销金额和部门 ID (2 分)
- 根据部门ID查询部门预算档案底表中的剩余额度 (1 分)
- 根据校验结果返回审批或驳回，并在审批意见中写明原因 (2 分)

代码

```
// 客开: laigiao [ERP出站 SSO 握手、验票和用户页面] 2026-09-29
@RestController
public class SsoController {

    3 个用法
    private final OaTicketClient oaTicketClient;

    4 个用法
    private final TicketCache ticketCache = new TicketCache();

    > public SsoController(OaTicketClient oaTicketClient) { this.oaTicketClient = oaTicketClient; }

    @GetMapping("/sso/handshake")
    > public ResponseEntity<String> handshake() { return ResponseEntity.ok(body: "ok"); }

    @GetMapping("/sso/v5ticket")
    7 public ResponseEntity<Void> receiveTicket(@RequestParam(value = "v5ticket", required = false) String v5ticket,
        @RequestParam(value = "ticket", required = false) String ticket) {
        String actualTicket = firstNonBlank(v5ticket, ticket);
        if (actualTicket == null) {
            return redirect(location: "/login?error=invalid-ticket");
        }
        try {
            OaTicketClient.TicketUser user = oaTicketClient.verify(actualTicket);
            if (user == null) {
                return redirect(location: "/login?error=invalid-ticket");
            }
            ticketCache.put(actualTicket, user.getLoginName(), user.getMemberId(), user.getMemberName());
            return redirect(location: "/login?v5ticket="
                + java.net.URLEncoder.encode(actualTicket, StandardCharsets.UTF_8.name()));
        } catch (Exception e) {
            return redirect(location: "/login?error=verify-failed");
        }
    }

    @GetMapping("/login")
    7 public ResponseEntity<String> login(@RequestParam(value = "v5ticket", required = false) String v5ticket,
        @RequestParam(value = "ticket", required = false) String ticket,
        @RequestParam(value = "error", required = false) String error) {
        String actualTicket = firstNonBlank(v5ticket, ticket);
        TicketCache.CachedUser user = ticketCache.get(actualTicket);
        if (user == null && actualTicket != null) {
            try {
                OaTicketClient.TicketUser verified = oaTicketClient.verify(actualTicket);
                if (verified != null) {
                    ticketCache.put(actualTicket, verified.getLoginName(), verified.getMemberId(), verified.getMemberName());
                    user = ticketCache.get(actualTicket);
                }
            } catch (Exception ignored) {
                // 页面只返回固定失败文案，避免泄露验证异常详情。
            }
        }
        if (user == null) {
```

```

0 // 客开: laiqiao [读取 OA 响应头中的 LoginName 等用户信息] 2026-09-29
1 2 个月法
2 Q~
3
4 public TicketUser verify(String ticket) throws Exception {
5     if (ticket == null || ticket.trim().length() == 0) {
6         return null;
7     }
8     String target = oaBaseUrl + "/thirdpartyController.do?ticket="
9         + java.net.URLEncoder.encode(ticket, StandardCharsets.UTF_8.name());
10    HttpURLConnection connection = null;
11    try {
12        connection = (HttpURLConnection) new URL(target).openConnection();
13        connection.setInstanceFollowRedirects(false);
14        connection.setRequestMethod("GET");
15        connection.setConnectTimeout(5000);
16        connection.setReadTimeout(5000);
17        connection.getResponseCode();
18        String loginName = connection.getHeaderField(name: "LoginName");
19        if (loginName == null || loginName.trim().length() == 0) {
20            return null;
21        }
22        String memberName = connection.getHeaderField(name: "MemberName");
23        if (memberName != null && memberName.trim().length() > 0) {
24            memberName = URLDecoder.decode(memberName, StandardCharsets.UTF_8.name());
25        }
26        return new TicketUser(loginName, connection.getHeaderField(name: "MemberId"), memberName);
27    } finally {
28        if (connection != null) {
29            connection.disconnect();
30        }
31    }
32 }
33
34 }
35
36 }

```

八、编写超级节点——预算校验

结果：

1. 自行设计费用报销单据，包含 部门预算档案底表、费用报销申请流程表（5 分）
2. 编写超级节点核心代码，继承 BaseSuperNodeAction，实现
 - 获取表单中的报销金额和部门 ID（2 分）
 - 根据部门ID查询部门预算档案底表中的剩余额度（1 分）
 根据校验结果返回审批或驳回，并在审批意见中写明原因

代码:

```

77 private SuperNodeResponse validateBudget(Map<String, Object> params) {
78     try {
79         ExpenseData expense = loadExpenseData(params);
80         BudgetDefinition budget = loadBudgetDefinition();
81         BigDecimal remaining = queryRemaining(budget.tableName, budget.departmentColumn,
82             budget.remainingColumn, expense.departmentId);
83         if (remaining == null) {
84             return human("未找到该部门的预算档案");
85         }
86         return budgetResponse(expense.amount, remaining);
87     } catch (BudgetValidationException e) {
88         return human(e.getMessage());
89     } catch (Exception e) {
90         LOG.error("message: 预算校验超级节点执行失败", e);
91         return human("预算校验异常, 请转人工处理");
92     }
93 }

```

// 客开: laiqiao [读取报销表单和金额、部门] 2026-09-30

1个月法

```

96 private ExpenseData loadExpenseData(Map<String, Object> params)
97     throws Exception {
98     if (cap4FormManager == null || params == null) {
99         throw new BudgetValidationException("预算校验依赖或流程参数缺失");
100     }
101
102     Map<?, ?> formData = asMap(params.get("CTP_FORM_DATA"));
103     FormBean expenseForm = formData == null ? null : (FormBean) formData.get("formBean");
104     Long masterId = toLong(params.get("WF_SUPER_NODE_MASTER_ID"));
105     if (expenseForm == null || masterId == null) {
106         throw new BudgetValidationException("无法获取费用报销表单数据");
107     }
108
109     FormDataMasterBean expenseData = cap4FormManager.getDataMasterBeanById(masterId, expenseForm, strings: null);
110     FormFieldBean amountField = expenseForm.getFieldBeanByDisplay(EXPENSE_AMOUNT_DISPLAY);
111     FormFieldBean departmentField = expenseForm.getFieldBeanByDisplay(EXPENSE_DEPARTMENT_DISPLAY);
112     if (expenseData == null || amountField == null || departmentField == null) {
113         throw new BudgetValidationException("费用报销表单缺少部门或报销金额字段");
114     }
115
116     BigDecimal amount = toDecimal(expenseData.getFieldValue(amountField.getName()));
117     Long departmentId = toLong(expenseData.getFieldValue(departmentField.getName()));
118     if (amount == null || departmentId == null) {
119         throw new BudgetValidationException("报销金额或部门值为空或格式不正确");
120     }
121     return new ExpenseData(amount, departmentId);
122 }

```

// 客开: laiqiao [读取部门预算底表定义] 2026-09-30

1个月法

```

25 private BudgetDefinition loadBudgetDefinition() throws Exception {
26     FormBean budgetForm = cap4FormManager.getFormByFormCode(BUDGET_FORM_CODE);
27     if (budgetForm == null || budgetForm.getMasterTableBean() == null) {
28         throw new BudgetValidationException("部门预算档案底表不存在");

```